

FIRST TERMINAL SAMPLE QUESTION - 2083

ROLL NO.		GRADE	XII
SUBJECT	PYTHON PROGRAMMING	FULL MARKS	25
DATE		TIME	1 Hrs.

Group 'A' "Answer in short" [5 * 1 = 5]

- What do you mean by inheritance?
- Explain polymorphism.
- Differentiate between list and tuple.
- What do you mean by file handling?
- How do you plan to use exception handling on your project?

Group "B" "Objective Questions" [5 *1 = 5]

- Which of the following methods is used to exclusively write a new file?
 - r
 - w
 - a
 - x
- List is mutable?
 - True
 - False
- If x = "Nepal", the output of print(x[:3]) is?
 - Nepal
 - Nep
 - Ne
 - Nepa
- A special string literal used in Python to document modules, functions, classes, and methods is known as _____.
- What will print(len("Samakhusi")) return?
 - 8
 - 9
 - 11
 - Error

Group C "Output of the code" [1*5 = 5]

- Provide the output of the following code.

```
class BankAccount:
    def __init__(self, balance):
        self.__balance = balance

    def show_balance(self):
        return self.__balance

    def withdraw(self, amount):
        try:
            if amount <= 0:
                raise ValueError("Withdrawal amount is negative.")

            if amount > self.__balance:
                raise ValueError("Insufficient balance.")

            self.__balance -= amount
            print("Withdrawal successful.")

        except (ValueError, TypeError) as error:
            print("Withdrawal error:", error)

account = BankAccount(1000)

try:
    account.withdraw(10000)
except ValueError:
    print("Please enter a valid numeric amount.")

print("Bank Balance:", account.show_balance())
```

Group D "WAP" [1*5 = 5]

- During heavy rainfall, Krishna Bhir in Nepal is highly vulnerable to landslides, which can block the highway and disrupt transportation. The Ministry of Transport plans to develop a Python-based landslide monitoring system and

has announced a vacancy for a Python Developer. As an applicant, you are required to design and develop the proposed system.

Create a Python class named **LandslideMonitor** with:

- Attributes: **location**, **rainfall**, and **road_status**
- A constructor to initialize these attributes
- A method named **display_warning()** that displays:
 - "Road Open: Low Landslide Risk" if rainfall is below 100 mm
 - "Road Closed: Landslide Warning" if rainfall is 100 mm or above
 -
- Create one object for **Krishna Bhir** with rainfall of **125 mm** and road status **"Blocked"**. Call the **display_warning()** method.

Group E “Debuging” [1*5 = 5]

1. The following Python program is intended to calculate the **molarity and molality of a 13% by-weight H₂SO₄ solution** having a density of **1.020 g/cm³**. However, the program contains errors. Identify the errors and rewrite the program correctly.

```
Molarity
1 percentage = 13
2 density = 1.020
3 molar_mass = 98
4
5 try
6 molarity = (density * percentage) / molar_mass
7 molality = (100 * percentage) / molar_mass * (100 - percentage)
8
9 print("Molarity =", round(molarity, 3), "mol/L")
10 print("Molality =", round(molality, 3), "mol/kg")
11
12 except ZeroDivisionError
13 print("The denominator cannot be zero")
```